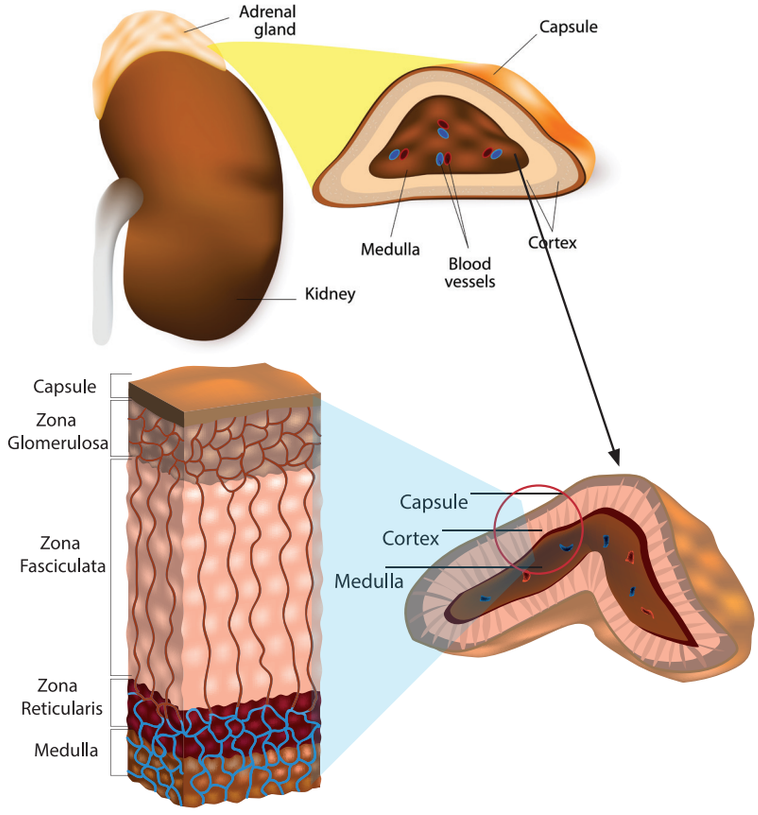


Figure 11.6 Structure of adrenal gland

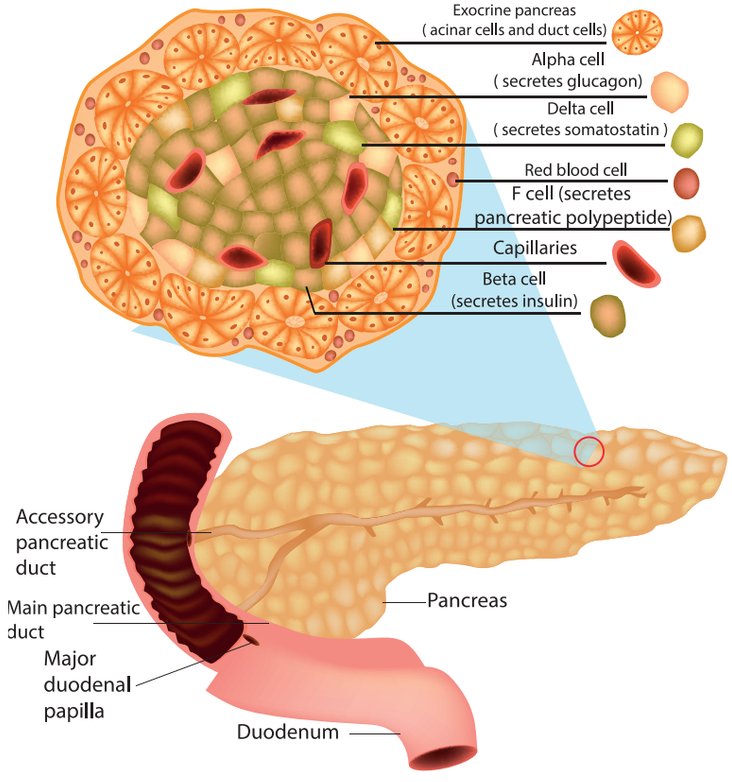


Figure 11.7 Structure of Islets of langerhans (pancreas)

11.2.8 Pancreas

Pancreas is a composite gland which performs both exocrine and endocrine functions. It is located just below the stomach as a leaf like structure. The pancreas is composed of two major tissues such the acini and islets of langerhans. Acini secretes digestive enzymes and the islets of langerhans secretes hormones like insulin and glucagon. Human pancreas has one to two million islets of langerhans. In each islet about 60% cells are beta cells, 30% cells are alpha cells and 10% cells are delta cells. The alpha cells secrete glucagon, the beta cells secrete insulin and delta cells secrete somatostatin.

Insulin: Insulin is a peptide hormone and plays an important role in glucose homeostasis. It's main effect is to lower blood glucose levels by increasing the uptake of glucose into the body cells, especially muscle and fat cells. Insulin also inhibits the breakdown of glycogen to glucose, the conversion of amino acids or fats to glucose, so insulin is rightly called a hypoglycemic hormone.

Glucagon: Glucagon is a polypeptide hormone. It is a potent hyperglycemic hormone that acts on the liver and promotes the breakdown of glycogen to glucose (Glygogenolysis), synthesis of glucose from lactic acid and from non-carbohydrate molecules (Gluconeogenesis). Glucagon releases glucose from the liver cells, increasing the

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